

A FOUR-ELEMENT COUNTEREXAMPLE TO CONJECTURE 6.1 OF MOHAN AND NEETU ON TWO DISJOINT ABELIAN SETS

ABSTRACT. Call a subset S of a group *abelian* when $\langle S \rangle$ is abelian. Conjecture 6.1 of Mohan and Neetu (arXiv:2607.11194v1, §6) asserts that if $S = A \cup B$ is a finite subset of a torsion-free group, A and B disjoint abelian sets, and $|S^2| \leq 3k - 3$, then $\langle S \rangle$ is abelian; the usage in that paper fixes $k = |S|$. The conjecture is false. In the Klein bottle group $\text{BS}(1, -1)$, written additively in two coordinates, the four-element set $S = A \cup B$ with $A = \{(-1, 1), (-1, 3)\}$ and $B = \{(1, 1), (1, 3)\}$ has A and B disjoint abelian sets, $|S^2| = 9 = 3|S| - 3$, and $\langle S \rangle$ nonabelian. This set is not new: it is the union of the first two of the three parts of Example 6.1 of that paper, printed two lines below the conjecture, at parameter value 2. No theorem of the paper is contradicted.

1. THE STATEMENT

Throughout, G is a group with identity e , $S \subseteq G$ is finite, and

$$S^2 = \{s_1 s_2 : s_1, s_2 \in S\}$$

is the product set over *ordered* pairs. Following [2, §1] — “a set S is said to be abelian if $\langle S \rangle$ is an abelian subgroup of G ” — a set is called *abelian* when the subgroup it generates is abelian. Freiman, Herzog, Longobardi and Maj proved that if G is *orderable*, $|S| \geq 3$ and $|S^2| \leq 3|S| - 3$, then S is abelian [1, Theorem 1.3], quoted as [2, Theorem 2.3]; the last section of [2], “Future Problems”, proposes ways to keep that conclusion while weakening the hypothesis. The first of them is:

Conjecture 6.1 (Mohan and Neetu). Let S be a nonempty finite subset of a torsion-free group G such that $S = A \cup B$, where A and B are disjoint abelian sets. If $|S^2| \leq 3k - 3$, $\langle S \rangle$ is abelian.

The conjecture does not define k ; the usage in [2] fixes $k = |S|$. Example 6.1, printed two lines below the conjecture and quoted in full in §2, asserts “ $|S^2| = 3|S| - 3$ ” of a set satisfying the antecedent, and [2, Theorem 2.9] writes “where $k = |S|$ ”. This is not a formality: under the alternative reading $k = |A|$ the set exhibited below would fall outside the antecedent, since there $3|A| - 3 = 3 < 9 = |S^2|$.

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2. THE WITNESS

Let $K = \mathbb{Z} \rtimes_{-1} \mathbb{Z} = \text{BS}(1, -1)$ be the Klein bottle group, in the coordinates used by [2, §5]: elements are pairs $(r, n) \in \mathbb{Z}^2$ and

$$(1) \quad (r, n)(s, m) = (r + (-1)^n s, n + m), \quad (r, n)^{-1} = (-(-1)^n r, -n),$$

with $e = (0, 0)$. This is the group and the operation of [2]: [2, §5] writes $\text{BS}(1, q) \cong \mathbb{Z}[1/q] \rtimes \mathbb{Z}$ with $(r, n)(s, m) = (r + q^n s, n + m)$, and the examples of its §6 are stated in a group the authors write $\mathbb{Z} \rtimes \mathbb{Z}$; that $q = -1$ there is fixed by their Example 6.2, which records the identity $(i, 1)(j, 1) = (i - j, 2)$, and for $q = 1$ the group would be abelian. All coordinates therefore stay in $\mathbb{Z}$ and every computation below is on pairs of integers.

Lemma 1. *K is torsion-free and nonabelian.*

Proof. By (1) the second coordinate is additive, so $(r, n)^j$ has second coordinate jn : if $n \neq 0$ and $j \neq 0$ this is nonzero, so $(r, n)^j \neq e$. If $n = 0$ then $(r, 0)^j = (jr, 0) \neq e$ whenever $r \neq 0$ and $j \neq 0$. Hence no element other than e has finite order. Finally $(-1, 1)(1, 1) = (-2, 2)$ while $(1, 1)(-1, 1) = (2, 2)$. $\square$

Lemma 2. *If the elements of a set $X \subseteq G$ commute pairwise, then $\langle X \rangle$ is abelian, i.e. X is an abelian set in the sense of [2].*

Proof. The centraliser $C_G(X)$ is a subgroup containing X , hence containing $\langle X \rangle$; so $\langle X \rangle \subseteq C_G(\langle X \rangle)$. $\square$

Two lines below Conjecture 6.1, the authors print:

Example 6.1 (Mohan and Neetu). Let

$$\begin{aligned} A &= \{(-1, 1), (-1, 3), \dots, (-1, 2k - 1)\}, \\ B &= \{(1, 1), (1, 3), \dots, (1, 2k - 1)\}, \\ C &= \{(-1, 2), (1, 2), (-1, 4), (1, 4), \dots, (-1, 2k), (1, 2k)\} \end{aligned}$$

be subsets of the group $\mathbb{Z} \rtimes \mathbb{Z}$. Let $S = A \cup B \cup C$. Then, we have $|S^2| = 3|S| - 3$. It is easy to observe that $\langle A \rangle$, $\langle B \rangle$, and $\langle C \rangle$ are abelian, whereas $\langle S \rangle$ is nonabelian.

Set the parameter the authors call k there — not the k of the conjecture — to 2, and delete the third part C . What is left is the following.

Theorem 3. *In K let*

$$A = \{(-1, 1), (-1, 3)\}, \quad B = \{(1, 1), (1, 3)\}, \quad S = A \cup B.$$

Then A and B are disjoint abelian sets, $|S| = 4$,

$$S^2 = \{(d, 2j) : d \in \{-2, 0, 2\}, j \in \{1, 2, 3\}\}, \quad |S^2| = 9 = 3|S| - 3,$$

and $\langle S \rangle$ is nonabelian. Since K is torsion-free by Lemma 1 and $9 \leq 3 \cdot 4 - 3$, Conjecture 6.1 of [2] is false.

Proof. The four listed pairs are distinct, and $A \cap B = \emptyset$ because the first coordinates -1 and 1 differ. Every second coordinate occurring in S is odd, so $(-1)^n = -1$ throughout and (1) gives, for $\varepsilon, \delta \in \{-1, 1\}$ and odd n, n' ,

$$(\varepsilon, n)(\delta, n') = (\varepsilon - \delta, n + n'), \quad \varepsilon - \delta \in \{-2, 0, 2\}.$$

Taking $\delta = \varepsilon$ gives $(0, n + n')$, which is symmetric in n, n' ; so the elements of A commute pairwise and so do those of B , and by Lemma 2 both are abelian sets.

For the product set: with $n, n' \in \{1, 3\}$ the sum $n + n'$ takes exactly the values $2, 4, 6$, and S contains both $(-1, n)$ and $(1, n)$ for $n = 1$ and $n = 3$, so at each of those three levels all four choices of (ε, δ) occur and $\varepsilon - \delta$ realises each of $-2, 0, 2$. That is the displayed set, and it has $3 \cdot 3 = 9$ elements, while $3|S| - 3 = 3 \cdot 4 - 3 = 9$.

Finally $(-1, 1)(1, 1) = (-2, 2)$ and $(1, 1)(-1, 1) = (2, 2)$, and both factors lie in S , so $\langle S \rangle$ is nonabelian. $\square$

No minimality is claimed: nothing here decides whether a witness of cardinality less than four exists, nor whether $3|S| - 3$ is the least doubling attainable by such a configuration, nor anything about host groups other than K .

Remark 4 (the contrast drawn after the conjecture). In [2] the conjecture is followed by “Whereas this may not hold for three abelian sets. Consider the following example.”, introducing Example 6.1. Read literally, the first sentence asserts only that the conclusion may fail for three abelian sets, which it does; but the contrast it draws with the two-set case is misleading, since by Theorem 3 the conclusion fails for two abelian sets as well, and the witness comes from the very example that sentence introduces.

Remark 5 (no theorem of [2] is contradicted). The set of Theorem 3 lies outside the hypotheses of the two positive results of [2] nearest to it. [2, Theorem 2.10] requires, among other hypotheses, $|A| \geq 4$ and $|B| \geq 3$, whereas here $|A| = |B| = 2$; and [2, Theorem 2.7] requires $|(A \cup B)^2| \leq 3|A \cup B| - 4 = 8$, whereas $|S^2| = 9$.

3. VERIFICATION

Every object used above is printed above, and the decisive computation is the sixteen ordered products of the four-element set S , carried out in the proof of Theorem 3 by the identity $(\varepsilon, n)(\delta, n') = (\varepsilon - \delta, n + n')$ for odd n, n' ; a reader needs no code.

The program `verify.py` accompanies this note and its recorded output is `verify.output.txt`. It re-implements (1) in exact integer arithmetic — the sign $(-1)^n$ taken as a parity lookup, so that no float can appear — forms the sixteen products of S one by one, compares S^2 with the nine pairs displayed in Theorem 3 *as a set of coordinate pairs, not merely by cardinality*, and confirms $|S^2| = 9 = 3|S| - 3$, that A and B are disjoint commuting pairs,

and that $(-1, 1)(1, 1) \neq (1, 1)(-1, 1)$. Torsion-freeness of K is confirmed there only on a bounded box of coordinates and exponents; the unbounded statement is Lemma 1. The recorded output also contains checks on objects not used above, among them the three-set Example 6.1 of [2] and the family of which the set of Theorem 3 is one member; nothing above depends on them.

REFERENCES

- [1] G. A. Freiman, M. Herzog, P. Longobardi, and M. Maj, *Small doubling in ordered groups*, J. Aust. Math. Soc. **96** (2014), 316–325. Its Theorem 1.3 is quoted as [2, Theorem 2.3], which is the form used here.
- [2] Mohan and Neetu, *On small doubling in right-ordered groups and Baumslag-Solitar groups – II*, arXiv:2607.11194v1, 13 July 2026. arXiv:2607.11194. Statement numbers are those the source compiles to.